

Microstrip Dipole Antenna Design for Direction of Arrival Estimation of Signal

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Abstract—In this research work, authors have done an effort to design, fabricate and experimental measure of microstrip dipole antenna (MDA) based on the microstrip planar technology. In order to estimate the direction of arrival (DOA) of incoming signal to any hand-held device, smart antenna is required, which can be replaced by proposed MDA along with softcomputing algorithm. This fact has been taken in mind before design of MDA and discussed. Different performance parameters of the antenna have been evaluated like reflection coefficient, impedance behaviors w.r.t frequency, Gain, Directivity, Efficiency, E and H plane radiation pattern, Main lobe direction (MLD), Half power beam width (HPBW), Front to back ration (FBR), and Side lob level etc. MDA has been fabricated in a laboratory and experimental measured for the few important parameters. Comparison has been done in order to validate the results. Close resemblance between both results validates the methodology.

Keywords—microstrip dipole antenna, direction of arrival, reflection coefficient, half power beamwidth, radiation pattern

I. INTRODUCTION

Direction of arrival (DOA) estimation of the incoming signal to any receiving device especially in wireless mobile communication application is generally done through the smart antenna. This is generally a group of antennas (array) associated with digital signal processing (DSP) chip. This led to a complex system to be implemented for DOA estimation. This smart system can be replaced by dipole antenna setup with softcomputing algorithm due to its radiation pattern dependence on the theta and phi.

In the published literature, it is found that many authors did work on smart antenna for DOA estimation [1]. However, no one has suggested this MDA approach to find DOA. Many authors attempted design of microstrip antenna in many ways. Microstrip line excited dipole antenna for two frequency application has been proposed to be applied for personal communication network (PCN) and wireless LAN (WLAN) having frequency 2.4 and 5.2 GHz using ADS and other software They analyzed it in terms of bandwidth, gain and reflection property, radiation pattern and half power beamwidth. [2-4]. Dipole antenna performance is dependent on the length, width and feed point location. Parametric analysis has been done to analyses performance in terms of resonance frequency and bandwidth and experimental verification has been done [5-6]. Microstrip dipole antennas for multiband application has been proposed by authors those are useful for radar detection, IMT-2000, WiFi, WLAN, Bluetooth, and ISM band respectively [7]. A wideband and dual-band dipole antenna with an integrated balun feed has been fabricated and experimentally tested at L and S-band validating calculations and measurements results [8]. For radio frequency identification (RFID) application, a triple-

band microstrip dipole tag antenna is designed using CST software [9]. Experimental measurements result of printed dipole antenna and inkjet printed textile antenna working at 2.4 GHz useful for wireless communication systems is presented [10-11]. A novel technique of direct-write dispenser printing on fabrics is explored, which has been applied to realize a dipole antenna for wearable electronic applications [12]. Micro strip dipole antenna arrays are designed for broadband 5G wireless cellular networks applications [13]. A printed dipole antenna is designed with two parasitic elements loaded by embedded resonators to provide reconfigurable patterns [14]. For the application of flexible display device, authors proposed microstrip printed antenna. Microstrip dipole antenna is developed using Kapton polyimide substrate (relative permittivity of 3.45 and a loss tangent of 0.002). Just varying length and width of the antenna in the UHF band, performance is evaluated [15]. Few authors extended the work just by presenting super wideband microstrip monopole antenna having human skull-shape. In that they combined various sizes of semicircular and rectangular-shaped radiating elements along with a microstrip feed line and a defected ground structure (DGS) [16]. A method of designing a filtering printed dipole antenna is proposed and analyzed in [17]. Researchers proposed idea of DOA estimation in complex way [18].

Looking in to the above published work, no authors have attempted and proposed individual dipole antenna possibility for DOA estimation. In this paper, MDA has been designed, developed and explored through its various performance parameters.

II. PROPOSED ANTENNA GEOMETRY

In order to utilize the dipole characteristics for finding the direction of arrival (DOA) of incoming signal to any handheld device, it is necessary to explore it thoroughly. Being commonly used, half wave dipole antenna design approach is proposed and designed using licensed CST software. In the proposed Microstrip Dipole Antenna (MDA), there are two microstrip leg ($l = \lambda/2$), which are fed from center as shown in Fig. 1. FR4, Epoxy glass (PTFE) is used to design and fabricate the antenna working at 2.4 GHz frequency. The dimensions of the MDA along with the feed is optimized using available mathematical relations as well as 'hit and trial'. Due to impedance mismatch high reflection may results, therefore exact location and dimensions of the feed is required to be optimized. Because dipole's radiation resistance is 73 ohms, which is very near the 50-ohm or 75-ohm characteristic impedances of microstrip feed line, its matching to the feed line is optimized especially at resonance. The geometrical parameters of the dipole, feed line and substrate are listed in Table 1. Since dipole radiation characteristics are dependent on the angle Theta/Phai (Fig. (1), Eq. (1), Eq. (2) and Eq. (3)), therefore its utility in the direction finding is feasible.

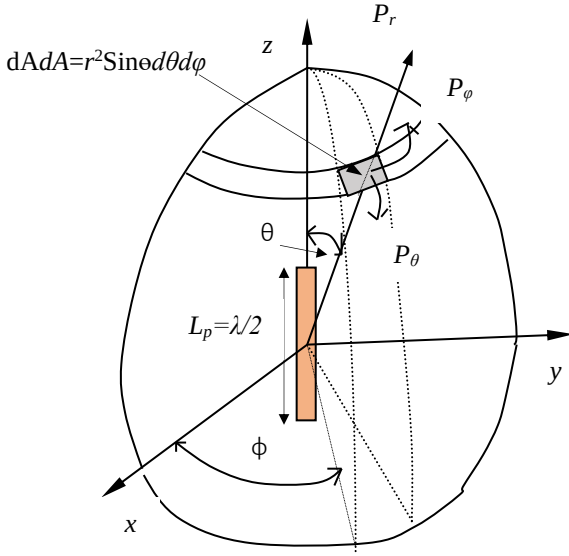


Fig.1. Dipole antenna 3 d view.

$$E_\theta = j\eta \frac{I_0}{2\pi r} e^{-jkr} \left[\frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right] \quad (1)$$

$$H_\phi \approx j \frac{I_0}{2\pi r} e^{-jkr} \left[\frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right] \quad (2)$$

$$P_{rad} = \eta \frac{|I_0|^2}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\cos^2\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} d\theta d\phi \quad (3)$$

$$U = \eta \frac{|I_0|^2}{8\pi^2} \sin^3\theta \quad (4)$$

$$D = 4\pi \frac{U_{max}}{P_{rad}} \approx 1.643 \quad (5)$$

$$Z_{in} = 73 + j42.5 \quad (6)$$

The electric and magnetic field components of a half-wavelength dipole are given as below Eq. (1), and Eq. (2). Also, the radiation intensity, U , and directivity, D is as blow Eq. (4), and Eq. (5) [18]. Since the imaginary part (reactance) associated with the input impedance of a dipole is a function of its length (for $l = \lambda/2$, it is equal to $j42.5$). Thus, the total input impedance for half wave dipole is given by Eq. (6). In above Equations, P_{rad} , I_0 , E_θ , H_ϕ , η , k is the radiated power, current flowing in dipole, electric and magnetic field components in θ and ϕ direction, intrinsic impedance, and wave number respectively. The input impedance of a MDA at resonance is real and at rest of the frequency points is dependent on the frequency, therefore the input power is independent of the feed-point along the length. The geometry of the proposed patch antenna is shown below (Fig. 2) and

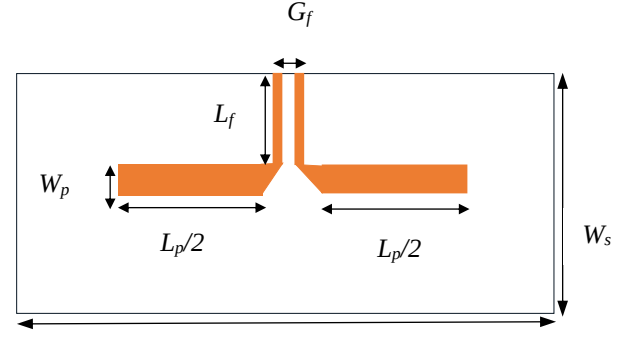


Fig. 2. Geometrical layout of the proposed

parameters of the antenna are listed in the Table 1 as below. Since dipole has been proposed and fabricated on dielectric without ground plane, therefore its radiation pattern is resemblance to simulated dipole and in is in all direction [19].

TABLE I. PROPOSED ANTENNA PARAMETERS WITH VALUES

S.N.	Parameters	Values
1.	Height of dielectric, h	1.6 mm
2.	Dielectric constant, ϵ_r	4.4
3.	Length of feed, L_f	7.94 mm
4.	Width of feed, W_f	0.32 mm
5.	Feed Gap, G_f	0.31 mm
6.	Substrate width, W_s	17.26 mm
7.	Substarete Length, L_s	37.88 mm
8.	Height of copper strip, t	0.07 mm
9.	Length of microstrip dipole, L_p	14.64 mm
10.	Width of microstrip dipole, L_w	1.38 mm

A. Parametric Analysis

For designing MDA working at 2.4 GHz resonance frequency, we have optimized the length, width and feed dimensions in such a way that it resonant within specified band frequency. In order to accomplish that, length of the dipole is changed to 13.5 mm, 14.5 and 15.5 mm. For each of the length, the simulated reflection coefficient plot is observed and plotted as shown in Fig. 3. In each case there is a need of appreciable dip in the resonance pattern at specified frequency point. For that high amount of the impedance matches required. Therefore, in each length, dimension of the feed is also calculated and optimized through 'hit and trial'. From the parametric plots of the various reflection coefficient as shown below (Fig. 3), it is observed that for higher the frequency range, length of the MDA needs to be reduced. Which is also validated by the relation between resonance frequency and length of antenna as $L_p = \lambda/2 = v/2f$ where v is the velocity of light ($= 3 \times 10^8$ m/s), f is frequency in Hz and L is length of dipole. In order to demonstrate the impedance matching and its behavior, smith impedance plot has been plotted for different radius of MDA i.e. 13.5 mm to 14.5 and 15.5 mm. and shown in Fig. 4. From the impedance chart, it is observed that region around resonance frequency in resonance graph below than -10 dB is well within the VSWR=2 circle.

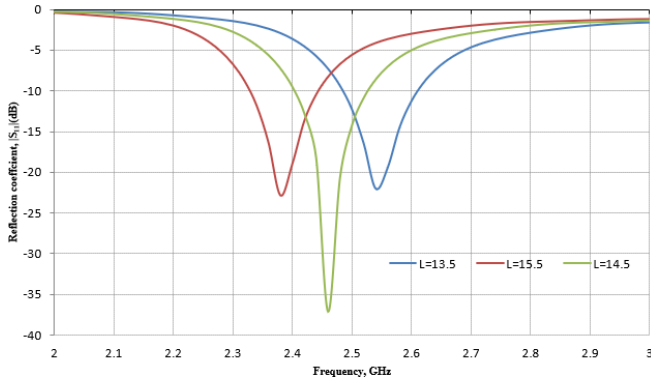


Fig. 3. Reflection coefficient for different length of MDA.

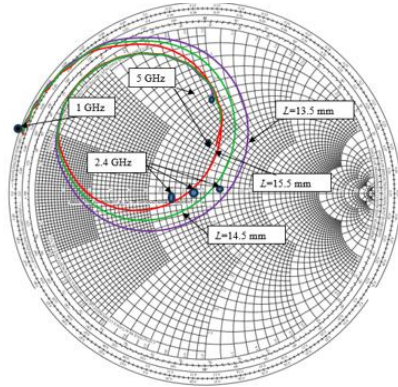


Fig.4. Smith impedance plot of different length MDA.

III. RESULTS AND DISCUSSION

In order to demonstrate the application of microstrip dipole antenna (MDA) for direction finding, its radiation characteristics has been explored. From the polar *E* and *H* plane radiation pattern (Fig.(7)), it is clear that its radiation characteristics is dependent on angle of theta (and Phi), which may locate the position of the observer. MDA has been proposed working at operational resonance frequency of 2.4 GHz. The reflection coefficient plot of the antenna working at 2.4 GHz frequency is shown below in Fig.5. Smith impedance plot of the MDA working at 2.4 GHz is shown in Fig. 6. From both plots, it can be observed that designed MDA is resonating almost at 2.4 GHz frequency. From the smith impedance chart, it can be seen the resonance plot section below -10 dBi, is well within VSWR=2 circle. Polar *E* and *H* plane directivity radiation pattern is shown in Fig.7. From the *E* and *H* plane polar radiation pattern, it is clear that its directivity is around 2.14 dBi. Since dipole efficiency is almost nearly 95 % therefore its gain is supposed to the near 1.64 (or 2.14 dBi).

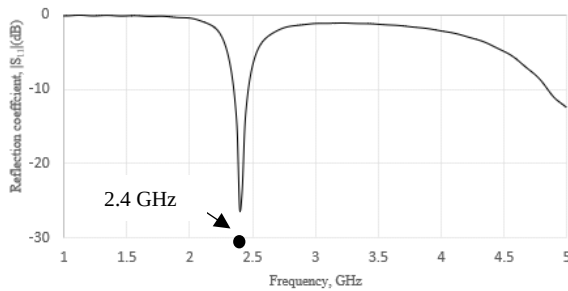


Fig.5. Simulated reflection coefficient plot of proposed MDA at 2.4 GHz.

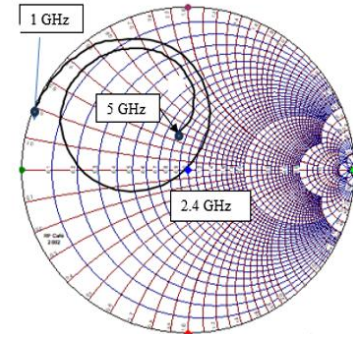


Fig.6. Smith impedance plot of MDA at 2.4 GHz, $L= 14.64$ mm.

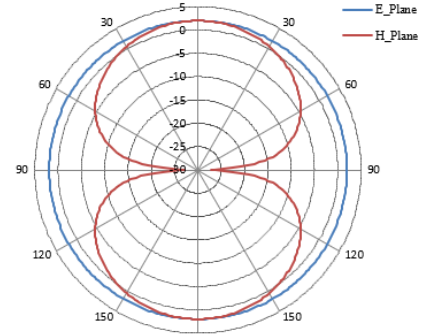


Fig. 7. E & H polar radition paterren of MDA.

The fabricated antenna prototype is shown in Fig. 8, whose experimental reflection coefficient plot (Fig. 9) shows the resonance nearly at 2.4 GHz. The numerical value of the most of the important performance parameters like $|S_{11}|$ (dB), VSWR, Gain, Directivity, Efficiency, MLD, HPBW, SLL etc. are listed as shown below in Table 2.



Fig. 8. Prototype of fabricated MDA at $2.4L=14.64$ mm



Fig. 9. Measured reflection coefficient plot for fabricated MDA at length, $L=14.64$ mm.

TABLE II. PERFORMANCE PARAMETERS OF PROPOSED MDA

S.N.	Parameters	Values
1.	Directivity, D (dBi)	1.98 dB
2.	Gain, G (dBi)	1.88 dB
3.	Ref. Coefficient, $ S_{11} $ (dB)	-27.2 dB
4.	Main Lobe Direction, MLD	$\sim 180^\circ$
5.	Side lobe level, SLL, dBi	-202.82
6.	Half Power Beamwidth, HPBW	85.66°
7.	Efficiency, η	96.8%
8.	Polarization, Axial Ratio	Linear
9.	Fron to back ratio, FBR	0.03235

IV. CONCLUSIONS

To comply with the need of DOA estimation of incoming signal in any handheld device, microstrip dipole antenna is may be useful due to its radiation pattern dependency on the 3-d angles. Therefore, MDA is proposed and optimized working at 2.4 GHz frequency band. CST simulator software has been used to design and develop MDA using readily available formula as well as ‘hit and trial’ method. Fabricated Antenna has been measured for the reflection coefficient, while other parameters like gain, directivity, efficiency, axial ration, MLD, HPBW, SLL, FBR etc. have been calculated through simulation. Parametric analysis has been done for various length of dipole antenna and plotted for impedance as well as reflection coefficient. On the basis of successful simulation results and optimized geometrical parameters, antenna has been fabricated in house in the RF and Microwave laboratory and tested experimentally using vector network analyzer. The simulation and experimental performance plots for reflection coefficient are compared to validate the results. Resemblance between both results shows that our methodology of MDA design is successful, as our designed, fabricated and experimentally measured antenna is resonating exactly at 2.4 GHz and providing desired results.

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